CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-503: Computer Networks

Session: December 2025 Examination | **Paper Code:** 18023 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) What is a MAC address? How is it different from an IP Address?

Comparison:

Feature	MAC Address (Physical Address)	IP Address (Logical Address)
Layer	Data Link Layer (Layer 2).	Network Layer (Layer 3).
Size & Format	48 bits (6 bytes), hex (e.g., `00:1A:2B:3C:4D:5E`).	32 bits (IPv4, e.g., `192.168.1.1`) or 128 bits (IPv6).
Nature	Hardcoded by manufacturer into NIC; globally fixed.	Dynamically/statically assigned based on network location.

Q2 (3M) Differentiate between a Hub and a Switch.

Differences:

- **Hub (Physical Layer - Layer 1):** Broadcast device that repeats every incoming signal across all ports, sharing a single collision domain. Half-duplex, no intelligence.
- **Switch (Data Link Layer - Layer 2):** Intelligent device maintaining a MAC Address Table to forward frames directly to designated destination ports. Dedicated collision domains per port, full-duplex.

Q3 (3M) What is Subnetting? Give an example.

Definition: Subnetting is the process of partitioning a single physical network into two or more smaller logical subnets by borrowing host bits to create subnet bits, reducing broadcast traffic and optimizing address allocation.
Example: Given Class C network `192.168.1.0/24` with default mask `255.255.255.0`. Borrowing 2 host bits gives subnet mask `255.255.255.192` (/26), creating 2^2 = 4 subnets with 62 usable hosts each:

- Subnet 1: `192.168.1.0` to `192.168.1.63`
- Subnet 2: `192.168.1.64` to `192.168.1.127`
- Subnet 3: `192.168.1.128` to `192.168.1.191`
- Subnet 4: `192.168.1.192` to `192.168.1.255`

Q4 (3M) What is CSMA/CD? How does it operate in Ethernet?

Carrier Sense Multiple Access with Collision Detection: A media access protocol used in shared half-duplex Ethernet:

1. **Carrier Sense:** Station listens before transmitting ("listen before talk"). If channel is idle, it begins transmitting.
2. **Collision Detection:** While transmitting, station monitors signal level. If collision occurs, it transmits a *jam signal* to notify all stations.
3. **Backoff:** Stations wait a random backoff period using the Binary Exponential Backoff algorithm before retransmitting.

Q5 (3M) **Differentiate between Synchronous and Asynchronous Protocols.**

Comparison:

- **Asynchronous Transmission:** Characters sent independently with start (0) and stop (1) framing bits. Clocks do not need synchronization. Inefficient for large high-speed data transfers.
- **Synchronous Transmission:** Continuous stream of data blocks transmitted synchronized to an external shared clock. Synchronized using sync character preambles (e.g., HDLC, PPP), achieving high throughput.

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) **Describe Multiplexing. Explain FDM, TDM, and WDM with diagrams.**

Definition: Combining multiple separate data streams into a single composite signal transmitted over a shared transmission medium.

Techniques:

- **FDM (Frequency Division Multiplexing):** Divides channel bandwidth into separate non-overlapping frequency bands with guard bands (Analog signals, radio, cable TV).
- **TDM (Time Division Multiplexing):** Interleaves time slots from multiple sources onto a high-speed digital carrier (Synchronous vs Statistical TDM).
- **WDM (Wavelength Division Multiplexing):** Multiplexes optical signals of differing wavelengths (colors) along a single optical fiber.

Q7 (7.5M) **Explain Error Detection Techniques: Parity Check, Checksum, and CRC (Cyclic Redundancy Check).**

Error Detection:

- **Parity Check:** Appends single parity bit (even or odd parity). Detects only odd numbers of bit errors.
- **Checksum:** Divides data into 16-bit words, adds them using 1's complement arithmetic, and appends inverted sum (used in TCP/IP).
- **Cyclic Redundancy Check (CRC):** Treats bit stream as polynomial coefficients and performs modulo-2 binary division by generator polynomial G(x). Appends remainder (FCS). Detects all single, double, odd, and burst errors up to generator length!

Q8 (7.5M) **Compare Circuit Switching, Packet Switching, and Message Switching.**

Comparison:

Feature	Circuit Switching	Packet Switching	Message Switching
Dedicated Path	Yes (setup before transfer).	No (packets routed dynamically).	No (store-and-forward per hop).
Bandwidth	Reserved/wasted if idle.	Dynamically shared on-demand.	Dynamically shared.
Delay	Setup delay high; transmission delay low.	Jitter/variable queuing delay.	High storage delay at each switch.
Modern Use	PSTN landline telephone.	Internet (IP networks).	Early telegraph, postal email.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **Explain the ISO/OSI Reference Model in detail. Describe the primary functions of each of the seven layers.**

Seven Layers of OSI Model:

1. **Physical Layer (Layer 1):** Transmission of raw unstructured bit streams over physical media (voltages, connectors, cables).
2. **Data Link Layer (Layer 2):** Framing, physical addressing (MAC), flow control, error control (CRC), and media access control.
3. **Network Layer (Layer 3):** Logical addressing (IP), routing (Dijkstra, distance vector), packet forwarding, and fragmentation.
4. **Transport Layer (Layer 4):** End-to-end process-to-process delivery (ports), segmentation and reassembly, connection control (TCP handshake), reliable flow control.
5. **Session Layer (Layer 5):** Dialogue control (simplex, half/full duplex), session checkpointing, synchronization tokens.
6. **Presentation Layer (Layer 6):** Data translation, character code conversion (ASCII/EBCDIC), compression, encryption/decryption (SSL/TLS).
7. **Application Layer (Layer 7):** Network virtual terminals, file transfer (FTP), web browsing (HTTP), email services (SMTP).

Q10 (15M) **Explain Sliding Window Protocols: Stop-and-Wait ARQ, Go-Back-N ARQ, and Selective Repeat ARQ with efficiency analysis.**

Flow and Error Control Protocols:

- **Stop-and-Wait ARQ:** Sender transmits 1 frame, starts timer, and waits for ACK before sending next. Efficiency: \eta = \frac{1}{1 + 2a} where a = T_p / T_t. Poor for high bandwidth-delay links.
- **Go-Back-N ARQ:** Sender window size W_s = 2^m - 1, receiver window W_r = 1. Sender can transmit multiple frames before receiving ACK. On loss, retransmits lost frame and all subsequent frames. Cumulative ACKs.
- **Selective Repeat ARQ:** Sender window W_s = 2^{m-1}, receiver window W_r = 2^{m-1}. Receiver accepts out-of-order frames within window. Sender retransmits *only the specific corrupted/timed-out frames* (highest throughput!).

Q11 (15M) **Explain Routing Algorithms: (a) Distance Vector Routing (Bellman-Ford) and (b) Link State Routing (Dijkstra). Discuss Count-to-Infinity problem.**

Routing Algorithms:

1. **Distance Vector Routing:** Each router maintains a routing table containing distance to each destination and next hop. Routers periodically share complete tables with immediate neighbors. Uses Bellman-Ford algorithm. Vulnerable to *Count-to-Infinity problem* when links fail (mitigated by Split Horizon and Poison Reverse).
2. **Link State Routing:** Each router floods Link State Packets (LSP) containing status of its local links to all routers in the entire autonomous system. Every router constructs a global topology map and computes shortest paths using Dijkstra's algorithm (used in OSPF). Fast convergence and no routing loops.

Q12 (15M) **Explain Congestion Control Mechanisms in Computer Networks: Leaky Bucket and Token Bucket Algorithms with comparative analysis.**

Traffic Shaping Algorithms:

- **Leaky Bucket:** Discharges data at a strictly constant, uniform output rate regardless of incoming burstiness. If incoming burst exceeds bucket buffer capacity, excess packets overflow and drop. Smooths traffic completely.
- **Token Bucket:** Generates tokens at a constant rate up to bucket capacity. Allows transmission of bursty traffic as long as sufficient tokens exist, but limits sustained rate. More flexible and widely used in modern QoS.

Q13 (15M) **Write detailed notes on: (a) DNS (Domain Name System), (b) Firewalls and Network Security, (c) SMTP and Email Architecture.**

Detailed Notes:

- **DNS:** Hierarchical distributed database mapping hostnames (`www.google.com`) to IP addresses (`142.250.190.46`) over UDP/TCP port 53.
- **Firewalls:** Security barrier controlling inbound/outbound network traffic based on rules. Types: Packet Filtering, Stateful Inspection, Proxy/Application-level gateways.
- **Email Architecture:** User Agent (UA) composes email, Mail Transfer Agent (MTA) transmits using SMTP (port 25/587), and Mail Access Agent (MAA) retrieves email to client via POP3 (port 110) or IMAP (port 143).
